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Să se determine formula de cuadratură cu grad maxim de exactitate:

$$\int_{-1}^1 x^2 f(x) dx = A_0 f(x_0) + A_1 f(x_1) + A_2 f(x_2) + R(f)$$

Este o formulă de tip Gauss cu $n=2 \rightarrow$ gradul maxim atins va fi $2 \cdot 2 + 1 = 5$

x_0, x_1, x_2 sunt rădăcinile polinomului ortogonal de grad 3 relativ la ponderea $w(x) = x^2$ pe intervalul $[-1, 1]$

$$P_0(x) = 1$$

$$P_1(x) = x + \alpha_1$$

$$\int_{-1}^1 x^2 P_0(x) P_1(x) dx = 0 \Leftrightarrow \int_{-1}^1 x^2 (x + \alpha_1) dx = 0$$

$$\Leftrightarrow \underbrace{\int_{-1}^1 x^3 dx}_{=0} + \alpha_1 \int_{-1}^1 x^2 dx = 0 \Leftrightarrow \alpha_1 = 0$$

$$\Rightarrow P_1(x) = x$$

$$P_2(x) = x^2 + \alpha_2 x + \beta_2$$

$$\left. \begin{aligned} \int_{-1}^1 x^2 P_0(x) P_2(x) dx &= 0 \\ \int_{-1}^1 x^2 P_1(x) P_2(x) dx &= 0 \end{aligned} \right\} \Leftrightarrow$$

$$\int_{-1}^1 x^2 \cdot 1 \cdot (x^2 + \alpha_2 x + \beta_2) dx = 0 \Leftrightarrow \int_{-1}^1 (x^4 + \alpha_2 x^3 + \beta_2 x^2) dx = 0$$

$$\Leftrightarrow \int_{-1}^1 x^4 dx + \alpha_2 \underbrace{\int_{-1}^1 x^3 dx}_{=0} + \beta_2 \int_{-1}^1 x^2 dx = 0$$

$$\frac{2}{5} + \beta_2 \frac{2}{3} = 0 \Rightarrow \beta_2 = -\frac{3}{5}$$

$$\int_{-1}^1 x^2 \cdot x \cdot (x^2 + \alpha_2 x + \beta_2) dx = 0 \Leftrightarrow \int_{-1}^1 (x^5 + \alpha_2 x^4 + \beta_2 x^3) dx = 0$$

$$\Leftrightarrow \underbrace{\int_{-1}^1 x^5 dx}_{=0} + \alpha_2 \int_{-1}^1 x^4 dx + \beta_2 \underbrace{\int_{-1}^1 x^3 dx}_{=0} = 0 \Rightarrow \alpha_2 = 0$$

$$\Rightarrow P_2(x) = x^2 - \frac{3}{5}$$

$$P_3(x) = x^3 + \alpha_3 x^2 + \beta_3 x + \gamma_3$$

$$\left\{ \begin{array}{l} \int_{-1}^1 x^2 P_0(x) P_3(x) dx = 0 \\ \int_{-1}^1 x^1 P_1(x) P_3(x) dx = 0 \\ \int_{-1}^1 P_2(x) P_3(x) dx = 0 \end{array} \right.$$

$$\left\{ \begin{array}{l} \int_{-1}^1 x^{2+1} (x^3 + \alpha_3 x^2 + \beta_3 x + \gamma_3) dx = 0 \\ \int_{-1}^1 x^2 \cdot x \cdot (x^3 + \alpha_3 x^2 + \beta_3 x + \gamma_3) dx = 0 \\ \int_{-1}^1 x^2 (x^2 - \frac{3}{5}) (x^3 + \alpha_3 x^2 + \beta_3 x + \gamma_3) dx = 0 \end{array} \right.$$

$$\underbrace{\int_{-1}^1 x^5 dx}_{=0} + \alpha_3 \underbrace{\int_{-1}^1 x^4 dx}_{=0} + \beta_3 \underbrace{\int_{-1}^1 x^3 dx}_{=0} + \gamma_3 \underbrace{\int_{-1}^1 x^2 dx}_{=0} = 0$$

$$\frac{2}{5} \alpha_3 + \frac{2}{3} \gamma_3 = 0$$

$$\int_{-1}^1 x^6 dx + \alpha_3 \underbrace{\int_{-1}^1 x^5 dx}_{=0} + \beta_3 \int_{-1}^1 x^4 dx + \gamma_3 \underbrace{\int_{-1}^1 x^3 dx}_{=0} = 0$$

$$\frac{2}{7} + \beta_3 \frac{2}{5} = 0 \Rightarrow \beta_3 = -\frac{5}{7}$$

$$\int_{-1}^1 (x^5 - \frac{3}{5} x^2) (x^3 + \alpha_3 x^2 + \beta_3 x + \gamma_3) dx = 0 \Leftrightarrow$$

$$\Leftrightarrow \int_{-1}^1 (\underbrace{x^8 - \frac{3}{5} x^5}_{=0}) dx + \alpha_3 \int_{-1}^1 (\underbrace{x^6 - \frac{3}{5} x^4}_{=0}) dx + \beta_3 \int_{-1}^1 (\underbrace{x^5 - \frac{3}{5} x^3}_{=0}) dx + \gamma_3 \int_{-1}^1 (x^4 - \frac{3}{5} x^2) dx = 0$$

$$\Leftrightarrow \alpha_3 \left(\frac{2}{7} - \frac{3}{5} \cdot \frac{2}{5} \right) + \gamma_3 \left(\frac{2}{5} - \frac{3}{5} \cdot \frac{2}{3} \right) = 0 \Rightarrow \alpha_3 = 0$$

Avem

$$\left\{ \begin{array}{l} \alpha_3 = 0 \\ \beta_3 = -\frac{5}{7} \\ \frac{2}{5} \alpha_3 + \frac{2}{3} \gamma_3 = 0 \Rightarrow \gamma_3 = 0 \end{array} \right. \Rightarrow P_3(x) = x^3 - \frac{5}{7} x$$

Rădăcinile lui $P_3(x)$ sunt $x_0 = -\sqrt{\frac{5}{7}}, x_1 = 0, x_2 = \sqrt{\frac{5}{7}}$

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Avem $R(P)=0 \quad \forall P \in \Pi_5$

Pt a determina A_0 considerăm

$$f = x(x - \sqrt{\frac{5}{7}}) \quad \begin{cases} f(0)=0 \\ f(\sqrt{\frac{5}{7}})=0 \end{cases}$$

$$\Rightarrow \int_{-1}^1 x^2 \cdot x \cdot (x - \sqrt{\frac{5}{7}}) dx = A_0 \cdot (1 - \sqrt{\frac{5}{7}}) \cdot (-2\sqrt{\frac{5}{7}})$$

$$\Leftrightarrow \int_{-1}^1 x^3 dx - \sqrt{\frac{5}{7}} \int_{-1}^1 x^2 dx = 2A_0 \frac{5}{7}$$

$\underbrace{\int_{-1}^1 x^3 dx}_{=0}$

$$\frac{2}{5} = 2A_0 \frac{5}{7} \rightarrow A_0 = \frac{7}{25}$$

Pt a determina A_1 considerăm

$$f(x) = (x + \sqrt{\frac{5}{7}})(x - \sqrt{\frac{5}{7}}) \quad \begin{cases} f(-\sqrt{\frac{5}{7}})=0 \\ f(\sqrt{\frac{5}{7}})=0 \end{cases}$$

$$\Rightarrow \int_{-1}^1 x^2 (x^2 - \frac{5}{7}) dx = A_1 \sqrt{\frac{5}{7}} (-\sqrt{\frac{5}{7}})$$

$$\int_0^1 x^3 dx - \frac{5}{7} \int_{-1}^1 x^2 dx = -A_1 \frac{5}{7}$$

$$\frac{2}{5} - \frac{5}{7} \cdot \frac{2}{3} = -A_1 \frac{5}{7} \Rightarrow A_1 = \frac{8}{45}$$

Pt a determina A_2 considerăm

$$f(x) = x(x + \sqrt{\frac{5}{7}}) \quad \begin{cases} f(0)=0 \\ f(-\sqrt{\frac{5}{7}})=0 \end{cases}$$

$$\int_{-1}^1 x^2 x (x + \sqrt{\frac{5}{7}}) dx = A_2 \sqrt{\frac{5}{7}} (2\sqrt{\frac{5}{7}})$$

$$\Leftrightarrow \int_{-1}^1 x^4 dx + \sqrt{\frac{5}{7}} \int_{-1}^1 x^3 dx = 2A_2 \frac{5}{7}$$

$\underbrace{\int_{-1}^1 x^3 dx}_{=0}$

$$\frac{2}{5} = 2A_2 \frac{5}{7} \rightarrow A_2 = \frac{7}{25}$$

$$\rightarrow \int_{-1}^1 x^2 f(x) dx = \frac{7}{25} f(-\sqrt{\frac{5}{7}}) + \frac{8}{45} f(0) + \frac{7}{25} f(\sqrt{\frac{5}{7}}) + R(f)$$